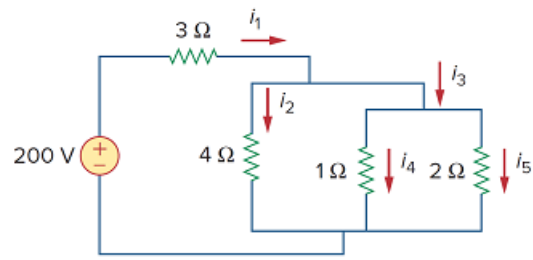


Problem

For the circuit in Fig. 2.95, determine i_1 to i_5 .

Figure 2.95:



Step-by-step solution

Step 1 of 9

Consider the following circuit diagram:

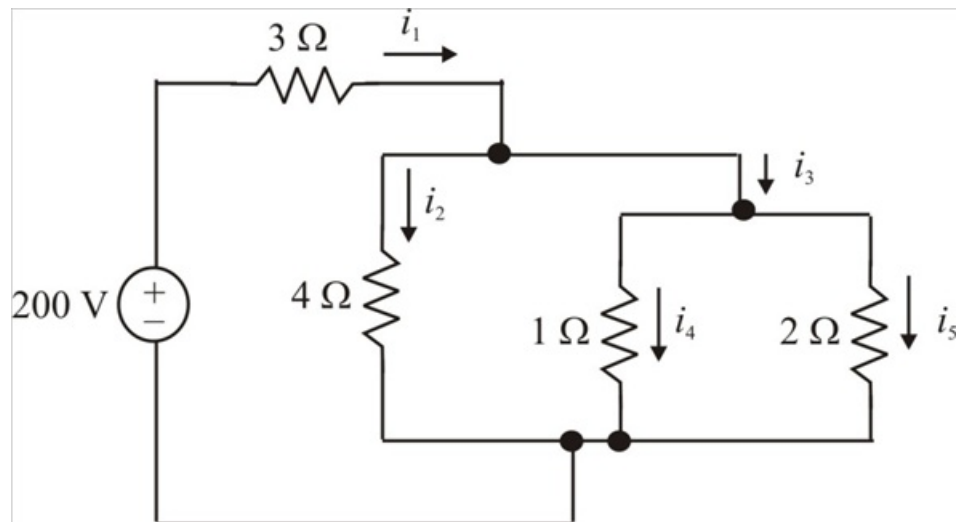


Figure 1

Step 2 of 9

To get R_{eq} , we combine the resistors in series and in parallel.

The 1 Ω and 2 Ω resistors are in parallel. Their combined resistance is,

$$1\Omega \parallel 2\Omega = \frac{1 \times 2}{1 + 2} = 0.66\Omega$$

Step 3 of 9

Then the reduced circuit is shown in Figure 2.

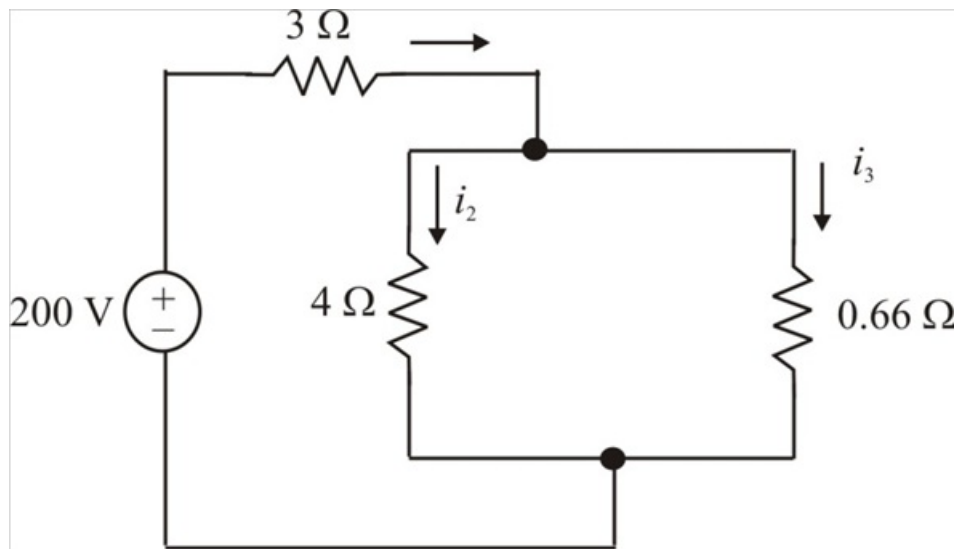


Figure 2

Step 4 of 9

The 0.66 Ω and 4 Ω resistors are in parallel. Their combined resistance is,

$$\begin{aligned} 0.66\,\Omega \parallel 4\,\Omega &= \frac{0.66 \times 4}{0.66 + 4} \\ &= 0.57\,\Omega \end{aligned}$$

Step 5 of 9

Then the reduced circuit is shown in Figure 3.

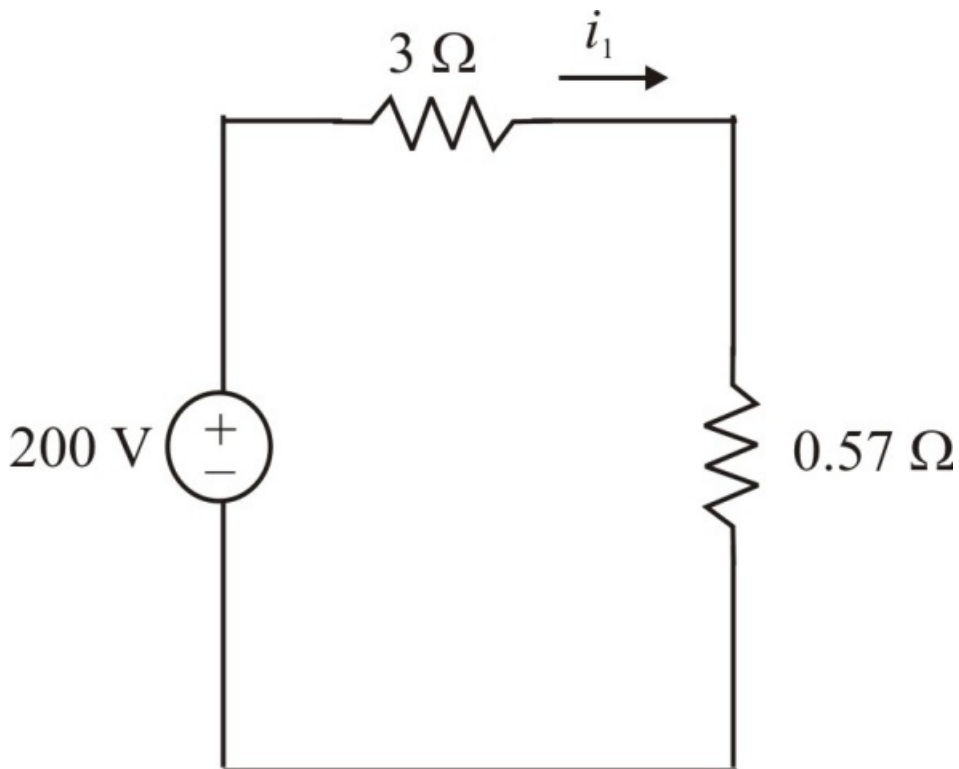


Figure 3

Step 6 of 9

The $0.57\ \Omega$ and $3\ \Omega$ resistors are in series. Their combined resistance is,

$$0.57\ \Omega + 3\ \Omega = 3.57\ \Omega$$

By using Ohms law, the current i_1 can be calculated as follows.

$$\begin{aligned} i_1 &= \frac{v}{R} \\ &= \frac{200}{3.57} \\ &= 56\ \text{A} \end{aligned}$$

Therefore, the current i_1 is $\boxed{56\ \text{A}}$.

Step 7 of 9

By using current division, the current i_2 can be calculated as follows.

$$\begin{aligned} i_2 &= \frac{0.66}{0.66 + 4} i_1 \\ &= \frac{0.66}{0.66 + 4} (56) \\ &= 8\ \text{A} \end{aligned}$$

Therefore, the current i_2 is $\boxed{8\ \text{A}}$.

Applying Kirchhoff's current law, the sum of currents entering any node is equal to the sum of currents leaving from that node. The current i_3 can be calculated as follows.

$$\begin{aligned} i_1 &= i_2 + i_3 \\ 56 &= 8 + i_3 \\ i_3 &= 48\ \text{A} \end{aligned}$$

Therefore, the current i_3 is $\boxed{48\ \text{A}}$.

Step 8 of 9

By using current division, the current i_4 can be calculated as follows.

$$\begin{aligned} i_4 &= \frac{2}{2 + 1} (i_3) \\ &= \frac{2}{2 + 1} (48) \\ &= 32\ \text{A} \end{aligned}$$

Therefore, the current i_4 is $\boxed{32\ \text{A}}$.

Step 9 of 9

Applying Kirchhoff's current law, the sum of currents entering any node is equal to the sum of currents leaving from that node. The current i_5 can be calculated as follows.

$$\begin{aligned} i_3 &= i_4 + i_5 \\ 48 &= 32 + i_5 \\ i_5 &= 16\ \text{A} \end{aligned}$$

Therefore, the current i_5 is $\boxed{16\ \text{A}}$.